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Review

Simulation of melt pool behaviour during additive manufacturing: Underlying physics and progress

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ABSTRACT

Reliable computational models of metal additive manufacturing will assist in optimising part quality, and are likely to play a role in component qualification. A key component of these models will be a detailed simulation of flow and heat transfer in and around the melt pool formed as the powder bed is melted. This paper reviews the burgeoning literature concerning melt pool simulation. The physical theory underlying the current benchmark models is first presented and the main approximations and assumptions discussed. The individual capabilities of the leading simulation groups around the world are listed in detail. Publications by less prominent research groups are also summarised. Finally, the overall status of melt pool simulation and the implications for model development are discussed.

1. Introduction

Metal additive manufacturing (AM) refers to the use of the energy provided by a laser, electron beam or electric arc to melt metal powder or wire, building a component over many passes, typically using computer-aided design (CAD) data. Processes are classified according to the form of the metal and the source of the energy. According to the terminology formulated by ASTM International, the relevant categories are powder-bed fusion and directed energy deposition [1]. Powder bed fusion can be divided into selective laser melting, in which regions of a metal powder bed are melted by a moving laser beam in an inert gas environment, and electron beam melting, in which the laser beam is replaced by an electron beam and operation occurs in a vacuum. In both cases, a roller or blade is used to spread additional powder layers over the bed between laser or electron beam passes. Directed energy deposition uses metal wire or powder, deposited or fed from above the component being built; these approaches are known respectively as wire-fed and blown-powder methods. The metal can be melted by a laser, electron beam or plasma arc; in the latter case, only wire-fed approaches have been widely used.

Simulation and modelling of AM processes are being pursued intensively by both researchers and by commercial software providers. The broad objectives are to accurately predict part properties and performance and to understand the sensitivities to important process parameters. As well as assisting in determining parameters for optimum part quality, it is proposed that reliable models can play a role in process qualification and part certification. A longer term aim is a simulation-based approach to actively control the AM production process, using feedback from process diagnostics.

A full simulation of an AM process can be broken down into a number of sub-models, covering residual stress, microstructure development, powder-bed raking for powder-bed AM or powder trajectories in blown-powder AM, and the detailed behaviour in and around the melt pool [2]. Ultimately, an important aim is to tie these sub-models into a single coupled ‘multi-scale’ simulation of the AM process, which together with an understanding of the structure–property relations, predicts part properties for given AM process parameters.

A number of useful reviews of AM modelling and simulation have been published in the past few years. King et al. [3] presented a comprehensive review of modelling at both powder and part scales, emphasising their work at Lawrence Livermore National Laboratory (LLNL), but also considering other contributions in some detail. A review of similar scope from the ESI group [4] is also valuable, but is less comprehensive in its survey of other work. Markl and Körner [5] presented a review that avoids any equations and emphasises powder-level approaches and multi-scale considerations. The team at Northwestern University has discussed the issues and difficulties in developing a multi-scale simulation [6,7], with a focus on the need for accurate process-structure-property relationships. A workshop held at Oak Ridge National Laboratory (ORNL) sought to identify the key simulation

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challenges [8]. Sames et al. [9] from ORNL provided a detailed description of AM metallurgy and processes, with a large section devoted to simulation methods. Schoinochoritis et al. [10] provided a good coverage of AM simulations up to 2014, including a handy table listing papers with validation data, but algorithm detail was limited to thermal stress. Karayagiz et al. [11] incorporated an extensive review of powder-bed fusion simulation as background before describing their own work. A 2012 review by Fan and Liou [12] is beginning to become dated, given the pace of AM simulation developments, but includes relevant sections on solidification and free surface methods. Gong et al. [13] provided a general description of the electron-beam melting process with particular attention to microstructure and a useful summary of reported modelling of the process. A NASA review [14] contains some less-cited references and an informative summary of microstructure evolution studies.

Our review is concerned primarily with modelling of the melt pool, and has two main aims that distinguish it from previous publications. The first is to provide an overview of the state-of the-art of relevant melt-pool modelling work in greater depth and detail than is possible in a broader review. The second is to assess the different approaches to incorporating the physical effects that arise in melt pool modelling. This includes a critical analysis of the different approaches used to treat important factors such as absorption of the incident laser or electron beam, metal vaporisation rate and heat transfer between particles. As well as AM, we draw upon findings from the related field of keyhole laser welding, for which the simulation understanding is more mature and free of powder complications. We expect this review will be of significant value to researchers developing or assessing melt pool models.

The review is focused on powder-bed, rather than blown-powder or wire-fed, processes. However, many of the problems that arise in powder-bed modelling are common to the other processes. In any case, the vast majority of the literature to date deals with powder-bed processes.

In Section 2, the relevant physics is described in some detail, including the more important of the approximations employed by the various research groups active in the field. In Section 3, the status of the work published by leading research groups is summarised. Section 4 discusses the current status of melt pool simulation and the implications for model development. Conclusions are presented in Section 5.

Some standard abbreviations employed throughout the review are listed here for convenience:

- Processes: EBM – electron beam melting, SLM – selective laser melting, DED – directed energy deposition, PBF – powder-bed fusion.
- Computational methods: FEM – finite-element method, FVM – finite-volume method, CFD – computational fluid dynamics, DEM – discrete-element method, LBM – Lattice-Boltzmann method, SPH – smoothed particle hydrodynamics.

2. Melt pool simulation theory

Models of the melt pool in powder-bed processes consider the processes occurring in the transient melt pool that forms where a laser or electron beam strikes a metal powder bed, and in the immediate vicinity (usually within about 1 mm) of the melt pool. As the beam passes by, it melts the powder, typically composed of ~20-50 μm metal particles, and creates a liquid melt pool that subsequently cools and solidifies, all of this occurring on a microsecond time scale.

While different computational methods have been applied in modelling the melt pool, solutions to the same fundamental set of physics equations are required, although the formulations used differ in detail. The formulation set out in the following sections is written from the perspective of computational fluid dynamics (CFD), which is the most widely used method for fluid flow modelling in general and for melt pool modelling in particular. Most CFD codes use either an FEM [15] or FVM [16] approach. Both discretise the model domain into a mesh of small elements or control volumes, respectively, for which the physics equations can be solved in simpler linearized forms before assembly to recreate the whole domain. Traditionally, FEM has been considered to have better handling of complex geometry and sometimes greater accuracy and FVM to be faster and more intuitive for coding new physics. In modern codes, however, these distinctions are very blurred and there is no clear favourite.

Two other computational methods, LBM [17] and SPH [18], have been applied to melt pool simulation. In LBM, rather than solving the Navier-Stokes equations directly, fictive fluid particles are created at a lattice of locations. Their subsequent collisions and movement are tracked and realistic fluid behaviour emerges. SPH is a mesh-free method in which the material is represented by a large set of fictive Lagrangian particles, each defined by a local kernel function. The standard physics equations are applied and decomposed into interactions between neighbouring 'particles'. There have been few applications of LBM and SPH to melt pool simulation to date. Commercial software is less mature and there are also difficulties in implementing some important physics such as temperature-dependent surface tension and complex boundary conditions. However, in the longer term, both methods show potential due to their implicit handling of complex interphase boundaries and their capacity to model the whole-body motion of solid metal particles.

2.1. Simulation requirements

The main requirements of the melt pool model are to:

- Predict the melt pool geometry and temperature distribution;
- Predict formation of defects such as porosity, spattering and other phenomena directly associated with the processes occurring in and close to the melt pool;
- Supply temperature data required to calculate the initial microstructure development near the melt pool;
- Supply an averaged melt pool temperature or heat source to a global model of the thermal and residual stress in the part;
- Provide mesoscale process understanding to inform the approximations made in the global model.

The simulation consists of a coupled solution of fluid flow and heat transfer that incorporates a number of physical effects (Fig. 1):

- Heating and melting of powder particles by the laser or electron beam;
- Surface tension and wetting effects within the partially melted powder bed and underlying substrate;
- Formation of a melt pool by coalescence of molten material;
- Flow within the melt pool generated by thermal buoyancy forces and the Marangoni effect associated with temperature-dependent surface tension;
- Creation, movement and freezing of porosity;
- Calculation of the liquid free-surface profile and its movement;
- Radiative and convective heat transfer from the metal surfaces;
- Evaporation of liquid, creating both recoil pressure on the liquid surface and another source of heat loss;
- Phase change of metal between liquid and solid with associated latent heat release or absorption.

2.2. Fundamental physical equations

All published CFD simulations of the melt pool region in AM start from the following set of simplifying approximations:

- The liquid pool is in a laminar flow regime;

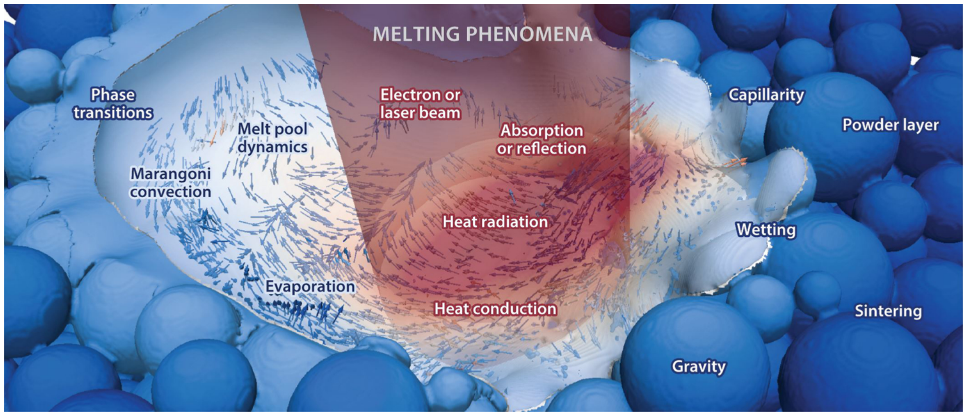

Fig. 1. Main physical processes in melt pool region during powder bed fusion (reproduced from Markl and Körner [5] with permission of ANNUAL REVIEWS; permission conveyed through Copyright Clearance Center, Inc).

- The liquid behaviour is Newtonian and furthermore has no temperature dependence, so the liquid has a constant dynamic viscosity μ;
- The liquid is incompressible with a constant density ρ. The volume expansion of the liquid is assumed sufficiently small that its only effect is to add a buoyancy force to the momentum equation. The volume thermal expansion coefficient β is usually taken to be constant.

The standard Navier–Stokes equations for mass and momentum conservation [19] then simplify to:

$$\nabla\cdot\boldsymbol{v} = 0 \tag{1}$$

$$\rho\frac{\partial\boldsymbol{v}}{\partial t} + \rho(\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = -\nabla P + \mu\nabla^2\boldsymbol{v} + \rho\boldsymbol{g} - \rho\boldsymbol{g}\beta(T - T_{\text{ref}}) + \boldsymbol{S_m} \tag{2}$$

Here P is pressure, $\boldsymbol{v}$ is the velocity vector and $\boldsymbol{g}$ is the acceleration due to gravity. The second-last term in eq. (2) is the buoyancy force for which T is the local temperature and T_{ref} is a reference temperature, usually set to the liquid melting temperature. The final term $\boldsymbol{S_m}$ represents any remaining momentum source terms and will be discussed in section 2.3.

Since the melt pool region has a free surface, an additional equation is needed to track the vapour interface. Most publications use a conventional volume-of-fluid approach in which the fluid volume fraction at a given point, F, obeys the following conservation equation:

$$\frac{\partial F}{\partial t} + \boldsymbol{v}\cdot\nabla F = 0 \tag{3}$$

The basic equation for energy conservation can be expressed in terms of enthalpy h as:

$$\rho\frac{\partial h}{\partial t} + \rho(\boldsymbol{v}\cdot\nabla)h = \nabla\cdot(k\nabla T) + S_h \tag{4}$$

Here k is the thermal conductivity and the final term S_h represents remaining enthalpy source terms as discussed in section 2.4.

For the phase change calculation, publications to date assign well-defined liquidus and solidus temperatures T_L and T_S to the metal and assume a linear variation of liquid fraction α between:

$$\alpha(T) = \begin{cases} 0 \text{ if } T < T_S \\ \dfrac{T - T_S}{T_L - T_S} \text{ if } T_S \le T \le T_L \\ 1 \text{ if } T > T_L \end{cases} \tag{5}$$

The temperature dependence of the enthalpy is conventionally expressed as the sum of an integral of specific heat $c(T)$ and a latent heat contribution. In the literature, the latent heat of melting L_m is linearly scaled by the liquid fraction:

$$h(T) = \int_0^T c(T)dT + \alpha(T)L_m \tag{6}$$

The other material properties used in the flow and thermal equations are temperature-dependent but generally straightforward. The exception is the thermal conductivity of the unconsolidated powder, for which a number of models have been employed, as will be reviewed in section 2.5.

The thermal boundary conditions applied by authors to the top surfaces of their models are discussed in section 2.4. On the other model surfaces a variety of conditions have been employed. Most commonly, the sides are made adiabatic, in view of the low conductivity of the loose powder, and the base is set to a fixed temperature. Pseudo-convective or semi-infinite solid approximations have also been used. The temperature fields are initialised to some uniform value, higher in the case of electron beam processes. At best, there is only cursory justification provided with an implicit assumption that the melt pool behaviour is insensitive to the boundary settings chosen.

2.3. Additional momentum source terms

In this section the additional momentum contributions represented by the final term S_m in eq. (2) are discussed.

2.3.1. Mushy zone flow resistance

In nearly all simulations reviewed, the effect of partial solidification on the metal flow is treated by a mushy zone method. In this approach the mushy zone ($0 < \alpha < 1$) is viewed as a porous medium with liquid and solid phases assumed to be in local equilibrium and exerting a drag force on the flow directly proportional to the local velocity. The constant of proportionality depends on the liquid fraction and the assumed form of permeability. Brent et al. [20] recommended a drag force in the form of the Carman–Kozeny equations and most authors have adopted this approach for AM melt pool simulations. The mushy zone contribution to the momentum source term becomes:

$$S_m(\text{mushy zone}) = -C[(1-\alpha)^2/(\alpha^3+\varepsilon)]\boldsymbol{v} \tag{7}$$

The constant C in eq. (7) is a value set large enough to ensure that velocity drops to zero as the local region freezes, typically 10^5 or greater. The particular value chosen for C can affect the solidification results, but this point has received little attention in the AM literature. An expression relating C to the viscosity and the dendrite cell size d_c of the microstructure,

$$C = 180\mu/d_c^2 \tag{8}$$

is used in modelling of casting and other processes [21]. The constant ε is an arbitrary small number incorporated simply to avoid division by zero, ideally as small as numerically possible and typically 10^{-5} or less.

2.3.2. Surface tension

The surface tension in the melt is believed to significantly affect the simulation results in several ways. Surface tension affects the curvature of the liquid interfaces and the shape of pores within the liquid. It leads to balling of the melt pool at large length-to-diameter ratios as described by the Plateau–Rayleigh instability theory. Capillary forces within the partially melted powder bed are thought to be important in the melt movement and coalescence. The Marangoni effect on the melt pool surface (variation of surface tension with temperature and/or species concentration) is one of the main drivers of the liquid flow.

The literature on AM simulation neglects any variation of surface tension with species concentration. The surface tension is approximated to vary linearly with temperature according to:

$$\sigma(T) = \sigma_{\text{ref}} + \frac{\partial\sigma}{\partial T}(T - T_{\text{ref}}) \tag{9}$$

Here, σ_{ref} is the surface tension at some reference temperature T_{ref} (typically the liquidus point) and $\partial\sigma/\partial T$ is the temperature derivative, which is taken to be a constant.

The contribution of surface tension to the momentum equation is expressed in a variety of mathematical forms in the literature according to the particular numerical scheme employed. The most fundamental expression is:

$$S_m(\text{surface tension}) = \sigma\kappa\boldsymbol{n} + \frac{\partial\sigma}{\partial T}[\nabla T - (\boldsymbol{n}\boldsymbol{\cdot}\nabla T)\boldsymbol{n}] \tag{10}$$

Here κ is the surface curvature and $\boldsymbol{n}$ is the (vector) outward normal to the surface. The first term in eq. (10) is the direct effect of surface tension acting normal to the surface and is often regarded as equivalent to an external pressure. The second term represents the Marangoni effect. Note that the portion inside the square brackets is the component of the temperature gradient parallel to the surface, so the entire term is effectively the spatial gradient of surface tension parallel to the surface.

Another feature of surface tension is the existence of an equilibrium contact angle between liquid and solid. This aspect has received scant attention in the literature but would seem to be quite important. For example, the area of contact between the initial melt and the substrate should affect the rate of cooling. Also, the contact angle between the edge of the melt pool and surrounding frozen material may affect the liquid surface profile, and influences the propensity to balling of the melt pool.

In the dynamic situation of a contact angle θ that differs from the equilibrium value θ_{eq}, a force is exerted on the fluid parallel to the wetted surface, as discussed in detail by Attar [22]. Another source term is added to the momentum equation:

$$S_m(\text{wetting}) = \sigma(\cos\theta - \cos\theta_{eq}) \tag{11}$$

The practical implementation of surface tension is complicated by the numerical difficulties in accurately defining normals and curvatures of surfaces, which are themselves approximations, based on a VOF interpolation, for example. Rather than attempting to directly prescribe the surface stresses, an equivalent volume force method is generally employed. Schemes are then required to reduce spuriously large velocities ('parasitic currents') in the vapour phase associated with the large density change across the liquid-vapour interface. Application of the contact angle is also difficult. Details of these aspects are beyond the scope of this review but can be explored in several publications [23–26].

2.3.3. Recoil pressure

A recoil pressure is exerted by the flux of atoms escaping as vapour from the liquid metal surface. Klassen et al. [27,28] gave a good description of the theory for electron-beam melting processes, which is summarised here. Consider an alloy composed of N elements $i = 1$ to N, with respective mole fractions χ^i. Atoms of each element vaporise at different rates, with the individual mass fluxes leaving the surface designated as $J_+{}^i$. Some of these atoms recondense, so the nett mass fluxes $J_{net}{}^i$ that *permanently* escape the surface are smaller quantities equal to $\varphi^i J_+{}^I$, where the factors $0 < \varphi^i < 1$ are known as the evaporation coefficients. In this multi-component system the nett mass flux of each element i is:

$$J_{net}^i = \varphi^i\chi^i\gamma_{act}^i P_s^i(T)\sqrt{\frac{m_A^i}{2\pi k_B T}} \tag{12}$$

Here T is the temperature of vapour leaving the surface (which in practice is taken to be the liquid surface temperature), k_B is Boltzmann's constant, $\gamma_{act}{}^i$ is the element's activity coefficient, $m_A{}^i$ is the atomic mass and $P_s{}^i(T)$ is the saturated vapour pressure of the pure component at this temperature. Klassen et al. [28] attempted some calculations of separate Al and Ti evaporation from Ti-6Al-4 V, but generally authors have approximated alloy evaporation during AM simulation as a pure material with effective physical properties. With this approximation the nett alloy mass flux simplifies to:

$$J_{net} = \varphi P_s(T)\sqrt{\frac{m_A}{2\pi k_B T}} \tag{13}$$

An alternative approach to the mass flux is based on kinetic theory (e.g. Plesset and Prosperetti [29]) and involves two 'accommodation' coefficients for evaporation from a saturated pressure P_s and for condensation from a far-field vapour pressure P_v. In the formulation used by Klassen et al., this complexity is subsumed in the single coefficient φ. The evaporation coefficient φ effectively incorporates an allowance for the accommodation coefficient for condensation and the ambient pressure, so is not necessarily a constant, as discussed later.

The saturation pressure at a surface temperature T can be calculated in terms of the known saturation pressure P_{atm} at the boiling temperature T_b. First the latent heat of vaporisation L_v is approximated to vary with temperature as:

$$L_v(T) = L_v(0)\sqrt{1 - \left(\frac{T}{T_c}\right)^2} \tag{14}$$

where T_c is the critical temperature. Then the Clausius–Clapeyron equation is integrated from T_b to T to obtain:

$$P_s(T) = P_{atm}\exp\left\{-\frac{L_v(0)m_A}{k_B T_c}\left[\frac{1}{T}\sqrt{(T_c^2 - T^2)} - \frac{1}{T_b}\sqrt{(T_c^2 - T_b^2)} + \arcsin\left(\frac{T}{T_c}\right) - \arcsin\left(\frac{T_b}{T_c}\right)\right]\right\} \tag{15}$$

This corresponds to eq. (5) of ref. [27], except that the sign of both arcsin terms has been reversed to correct an error in that article. The sign error is in fact unimportant, because in the AM literature an implicit assumption is made that T and T_b are both so small compared to T_c that the arcsin terms can be dropped and the square roots replaced by T_c. Then eq. (15) simplifies to:

$$P_s(T) \cong P_{atm}\exp\left[\frac{L_v(0)m_A(T - T_b)}{k_B T T_b}\right] \tag{16}$$

Some checks of these assumptions based on property data for Ti-6Al-4 V indicate the associated over-estimation of saturation pressure is small, about 5% for a surface temperature 100 K greater than T_b for example, and less at lower temperatures.

The recoil pressure P_{rec} depends not just on the saturated pressure $P_s(T)$ but also on the condensation rate. Klassen et al. [27] derive an expression for P_{rec} in terms of the evaporation coefficient and the pressure P_{Kn} on the outer side of the Knudsen vapour layer adjacent to the surface:

$$P_{rec} = \frac{1}{2}P_s + \frac{1}{2}(1 - \phi)\left(\frac{1}{2}P_{Kn} + \frac{1}{2}P_s\right) \tag{17}$$

They show that for electron beam melting processes, the evaporation coefficient φ is close to 0.82 and that the recoil pressure approaches a limiting (minimum) value of 0.56 P_s. Then from eq. (16) we have:

$$P_{rec}(T) \cong 0.56 P_{atm} \exp\left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b}\right] \tag{18}$$

The recoil pressure factor may be significantly larger than 0.56 in non-vacuum conditions because the evaporation coefficient φ depends on the Mach number of the vapour at the Knudsen boundary [28]. On the other hand, Semak and Matsunawa [30] quote a private communication from Anisimov that the recoil coefficient does remain close to the minimal value for 'practical values of the ambient pressure'. Eq. (18) has been used in simulations of keyhole laser welding with factors of 0.54 [31–34], 0.55 [35] and 0.6 [36], and with a factor of 0.54 in simulations of laser powder bed fusion [37–41].

To explain observed variations in laser penetration depth with the ambient pressure P_{amb}, Pang et al. [42] recommend that a minimum value of P_{amb} should be imposed on the recoil pressure obtained from eq. (18). They argue for this on the basis of the increased condensation rate caused by the confinement of vapour by an ambient pressure (which is known to increase the recoil pressure factor from 0.56 toward the upper limit of 1) and show that their modification improves the match of their keyhole welding simulations with experiment. However, the authors do not address the issue that their approach implies a recoil pressure would still be present when melt vaporisation is negligible.

2.4. Additional heat transfer source terms

In this section the additional heat transfer contributions represented by the final term S_h in eq. (4) are discussed, all being applied to the top surface.

2.4.1. Evaporative cooling

Evaporation of material from the liquid surface removes significant heat from the melt pool. In section 2.3.3, the rate of mass loss was discussed. Combining eq. (13) and (16), the nett mass flux from the surface, i.e. the balance of evaporation and condensation, becomes:

$$J_{net} = \varphi P_{atm} \sqrt{\frac{m_A}{2\pi k_B T}}\, exp\left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b}\right] \tag{19}$$

Note that the evaporation process for alloys is again approximated here as if from a pure material using effective physical properties.

The rate of enthalpy removal from the liquid by evaporation is equal to the mass flux multiplied by the specific enthalpy at the liquid surface. The enthalpy is implicitly removed with the vapour in Lattice-Boltzmann simulations [27,28]. However, in the finite element and finite volume methods used for most melt pool simulations, mass is not explicitly removed from the liquid domain so for correct temperature calculation only the latent heat of vaporisation is applied. The heat loss contribution to the energy equation is thus:

$$S_h(\text{evaporative cooling}) = \varphi L_v(T) P_{atm} \sqrt{\frac{m_A}{2\pi k_B T}}\, exp\left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b}\right] \tag{20}$$

Eq. (20) incorporates an allowance for condensation via the evaporation coefficient φ. For a vacuum process like EBM, the appropriate value of φ is 0.82 [27]. For laser heating, φ depends on the vapour Mach number M at the edge of the Knudsen zone [27] and calculations by Knight [43] show that φ varies as approximately $0.82\sqrt{M}$ for subsonic flow. However, where mentioned in laser welding and PBF simulations to date [32,33,37,38] a value of 0.82 has been applied, without explanation.

Most publications that include evaporative cooling have used a formulation similar to that above. However, there have been some variations.

Vasquez et al. [44] calculated vapour mass losses during DED by dividing the evaporative heat loss by L_v. The unusual aspect of their simulation was that, rather than applying evaporative heat loss as a boundary condition, they obtained it from the surface temperature gradient and the energy balance required to maintain a steady state. The details are not fully explained but it is inferred that the melt surface was fixed at T_b with an implicit assumption that evaporation is sufficient to suppress any superheating.

Heeling et al. [40] describe a unique method to incorporate evaporative heat loss. Their SLM simulation code considers both fluid and vapour cells in the domain. At each timestep, the next temperature T^n at a melt surface cell is calculated in the normal manner but *without* any explicit evaporation term, and the new recoil pressure P^n_{rec} is determined using eq. (18). Then the hypothetical temperature T that would produce a saturated vapour pressure equal to P^n_{rec} is found from eq. (16). (The justification for this step is not explained.) The difference (T - T^n) is said to represent the evaporative heat loss during the timestep and energy is conserved by converting a corresponding quantity of fluid to vapour in that cell.

2.4.2. Radiative cooling

Most publications include a standard grey-body radiative cooling term in the energy equation:

$$S_h(\text{radiative cooling}) = \sigma\varepsilon(T^4 - T_a^4) \tag{21}$$

Here σ is the Stefan–Boltzmann constant, ε is the surface emissivity and T_a is an ambient temperature, typically set to room temperature. This expression appears to be used throughout the domain with single emissivity and ambient temperature values, even when individual powder particles are simulated. The expression doesn't take into account interception of the emitted radiation by particles in the line of sight, and would therefore overestimate radiative cooling in the powder bed, and around porosity.

2.4.3. Convective cooling

A convective cooling term is usually included in the energy equation:

$$S_h(\text{convective cooling}) = h_c(T - T_a) \tag{22}$$

where h_c is a convective heat transfer coefficient and T_a is an ambient temperature, typically set to room temperature. A range of values for h_c have appeared in the literature; 10 W m^{-2} K^{-1} [12,41,45,46], 15 [47], 20 [48], 25 [49], 40 [50], 50 [51,52], 80 [37,53,54] and 100 [55,56]. There is little or no explanation, except that most of the larger values were intended to compensate for the omission of radiation [54–56]. In any case, convective cooling has much less influence than radiative cooling on the melt pool, and of course none for EBM processes.

2.4.4. Beam heating

The absorption of heat from the laser or electron beam is central to the AM process. Given this, it is interesting and perhaps concerning that many different approaches have been published. To describe the heating methods, it is helpful to first divide the models into two categories; those that model the powder bed as a continuum and those that consider individual particles.

Powder modelled as a continuum

For continuum models, all authors have applied a spatially-varying heat flux with a Gaussian beam cross-section. From that common starting point there are many variants. To describe them more easily, it is helpful to define a generic equation that encompasses the reported heat flux distributions:

$$q(r, z) = \frac{g\eta Q}{\pi R^2(z)} \exp\left[-\frac{gr^2}{R^2(z)}\right] f(z) \tag{23}$$

Here $q(r,z)$ represents a heat flux at radial distance r from the beam axis

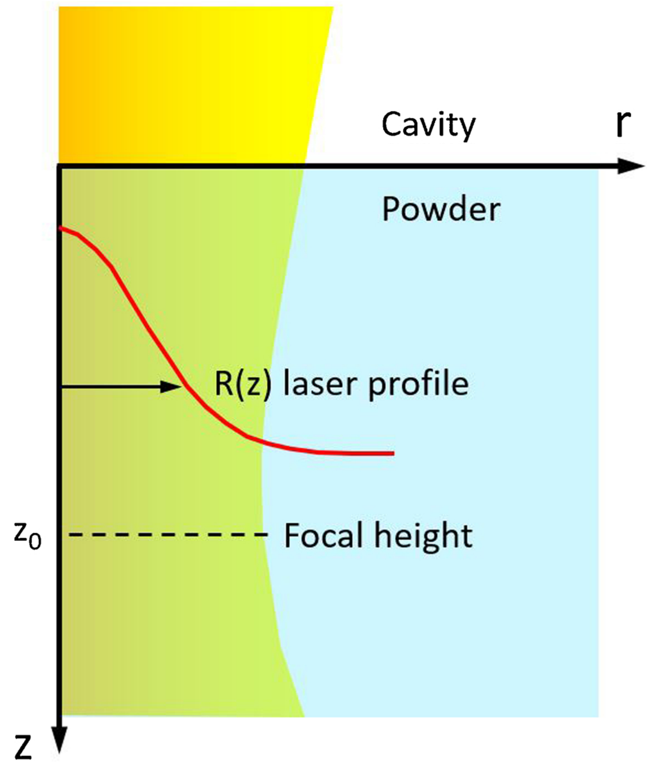

Fig. 2. Schematic of tapered laser beam with Gaussian profile (half-section).

and at depth z from the top of the powder bed. Fig. 2 illustrates the geometry.

The quantity Q is a measure of beam power, such that the area integral of eq. (23) over r should equal the total beam power. The meanings of the other parameters are as follows:

- The quantity $R(z)$ is a representative radius that defines the beam's Gaussian intensity profile at depth z. In virtually all studies, R is treated as a constant, i.e. a collimated beam is assumed. Just two groups [31,57] have modelled a diverging beam by setting $R(z) = R_0 + a|z_0 - z|$, where z_0 is the beam focus location and a is the beam divergence.
- The quantity g is an associated geometric factor that completes the Gaussian profile definition. For laser heating, $g = 2$ has normally been employed as expected for a Gaussian [37,41,45,48,50,54,57–68]. However, Karayagiz et al. [11] used $g = 0.5$, two groups [31,33,34] used $g = 3$, and one [69] omitted squaring the first R and had non-matching g values. One could argue that the factor g depends on the definition of R, but many of the papers do not indicate that this match has been made. For electron beam heating, the radial profile and hence g depend on the filament and deflection coil conditions [70]; two different groups [71,72] have both used $g = 4\ln(10)$.
- The quantity η refers to the material's absorptivity coefficient, i.e. the fraction of the incident radiation that is absorbed rather than reflected. It should be considered in tandem with the function $f(z)$, which defines how the power declines with distance travelled into the bed. Through ray tracing studies, Boley et al. [73] showed how multiple reflections increase the overall absorption of a powder bed compared to a flat surface, by a factor of 1.7 in the case of titanium and larger factors for more reflective materials.

The absorptivity of a given powder material is known to vary with a number of factors such as the nature of the particle surfaces, level of oxidation and laser wavelength. More significantly in terms of simulation methods, the absorptivity also varies with temperature; in particular the absorptivity decreases markedly as the surface starts to melt [62]. No allowance for this behaviour appears to have been made by any group, both for continuum and discrete powder models. The absorptivity is always regarded as being independent of temperature.

There is a variety of implementations of $f(z)$ in the literature. Most common is to omit any depth dependence $f(z)$ and apply the entire heat flux to the uppermost elements [11,41,44,48,54,58–62,65], with an implied absorptivity coefficient. Other authors prescribe a depth profile but there is little commonality in formulation:

- Mukherjee et al. [74] applied the general Beer–Lambert form of decay through the powder bed, i.e. $f(z) = \exp(-\beta z)$ where β is an optical extinction coefficient. (Strictly, though, the Beer–Lambert law applies to solid material, not a powder.) Additionally, they applied a heat flux to the top of the substrate that was determined by a separate calculation. Yadroitsava et al. [46] also employed an exponential decay.
- Two groups [67,68,75] effectively used $f(z) = 1 - z/s$, where s is a 'penetration depth' at which beam intensity has dropped to less than 1%. This form seems to stem from some EBM measurements [76] although Ladani et al. [68,75] were actually modelling laser heating.
- For EBM, Zah and Lutzmann [72] use $f(z) = 1 + 2z/s - 3z^2/s^2$, which has the problem that it increases initially with depth z. Their equation is sourced from an ARCAM web site which in turn relies on doctoral experimental work by Heinrich in 1973 (see ref. 16 in [72]). Chou and co-workers [45,63,64] also adopted this equation. Jamshidinia et al. [71] referenced Zah and Lutzmann but employed $f(z) = 1 - 2z/s - 3z^2/s^2$, which is problematic since it falls below zero for z much less than s.
- Gusarov et al. [77] applied depth and radial dependences derived from a complex two-flux laser radiation model. The equations can be a good approximation to a full ray-tracing calculation (refer Appendix A) but are lengthy and are so not reproduced here. Their method was adopted by two other groups [40,66].

2.4.5. Powder modelled as discrete particles

There has been a clear shift in recent years from treating the powder bed as a continuum to resolving individual particles. The current section examines how beam heating has been applied to individual particles in melt pool simulations. There have also been many related studies which have focused on determining only the absorption profile through analytic or ray-based methods, without looking at the subsequent melt pool formation. A brief summary of some of those publications is given in Appendix A.

The group headed by Körner at the University of Erlangen-Nuremberg has used Lattice-Boltzmann numerical models to study the melt pool, as distinct from the FEM and FVM models considered in the rest of this section.

- Attar [22] used a 2D LBM. She assumed radiation absorption of a Beer–Lambert form. The incident radiation intensity was given a Gaussian profile, with no depth dependence. Physical particles were modelled, composed of sub-particles by the underlying LBM treatment. The formulation is not made entirely clear, but from the absorption parameters given it would appear that effectively all radiation must have been absorbed by each physical particle encountered, i.e. no transmittance through particles was considered. No reflections were considered.
- In later work by the Erlangen group [78], analytic and experimental models of electron absorption and scattering in a powder bed were assimilated to arrive at (highly complex) expressions for the electron 'dose' vs depth. These dose curve calculations were subsequently applied in LBM models of EBM [27,28] as prescribed heating distributions.

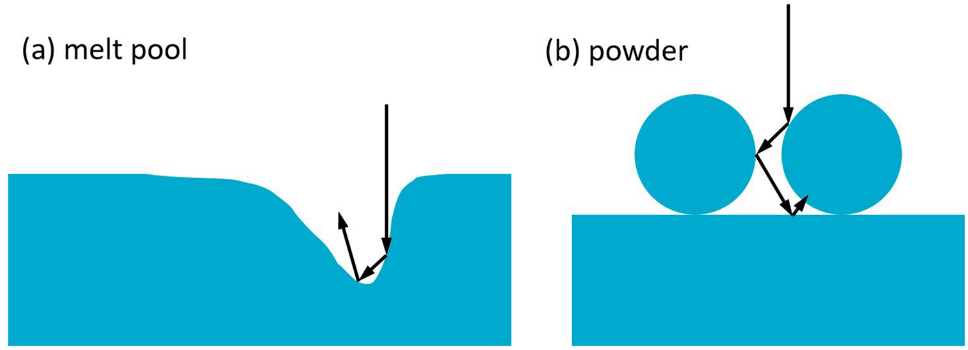

Fig. 3. Examples of enhanced absorption by multiple reflections from (a) melt pool surface and (b) powder particles.

Other authors have used FVM and FEM simulation methods. The heating methods can broadly be sorted in increasing complexity from prescribed profiles, to vertical rays with no reflection, to full ray tracing calculations (Fig. 3):

- Ganeriwala and Zohdi [50] used a combined DEM and finite difference approach; DEM particles for the powder layer and finite difference for the substrate. In the powder bed, each particle was given a lumped heat capacity. Laser power was applied to each particle based on its projected area and a Gaussian power profile as in eq. (23) with an absorptivity η and a depth profile $f(z) = \exp(-\beta z)$. This method is effectively a prescribed heating profile. King et al. [79] also applied an analytic heating profile, but a more sophisticated one, being derived from an analytic radiation model of powder heating [77].
- Lee and Zhang [37] applied a heat flux 'to the top free surface of the powder layer' according to a Gaussian power profile and absorptivity η, as per eq. (23) but with no depth function. It is presumed that by 'top free surface' they meant the unshadowed particle surfaces since the powder was meshed as particles derived from a DEM calculation. Several other groups [3,38,56,57,80] have used this approach, which can be regarded as ray tracing but without reflections.
- A ray-tracing method has been used for power absorption in several simulations of keyhole-mode laser welding [31,35,36,81], which is not a powder bed process but still relevant. The calculations started with a set of rays whose individual power densities were derived from a Gaussian beam intensity profile. Reflections were treated as specular and the Fresnel equation used to calculate the angle-dependent reflectivity R (and therefore absorptivity $1 - R$ for heating). Each ray was tracked through up to about 15 reflections, until the remaining ray intensity was negligible. Otto et al. [26] also used a ray-tracing method for laser keyhole welding but with two enhancements. One was that the full Fresnel equations were used to separate each ray into parallel and perpendicular polarisations with different reflectivities. The second is a suggestion in their wording that they applied the Gaussian power profile by grading the spatial distribution of the initial rays, which would be more computationally efficient than setting different ray energies.
- Gurtler et al. [69] seem to have been the first to use full ray-tracing for particle heating in AM melt-pool simulation. Like Otto et al., they assumed specular reflections and applied polarisation-dependent Fresnel equations to calculate reflectivities. Within each particle the transmittance was assumed to follow the Beer–Lambert law. Similar ray tracing may have also been applied by Matthews et al. [39], but their description is ambiguous and they may have instead simply applied profile expressions fitted from an earlier detailed ray-tracing study [82].
- Megahed et al. [4] applied a finite-volume formulation of a discrete ordinate radiation model. The radiation transfer equation is integrated over space by dividing directions into a finite number of discrete solid angles. A key advantage over ray tracing is allowing diffuse reflections. The incorporation into a forerunner of the ESI code has been described by Vaidya [83].

It is important to note that heating a powder bed with reflecting rays is fundamentally different to mimicking those reflections with a prescribed depth profile. A ray-based method is more realistic in that heat deposition is concentrated at the surfaces of particles rather than distributed through their volumes. This means that the surface melting, which greatly enhances heat transfer between particles, is established significantly earlier.

2.5. *Effective thermal conductivity of powder*

The thermal properties of the powder bed are central to the melt pool development, in particular the thermal conductivity. The background theory and reported methods of implementation are outlined for continuum and discrete models.

2.5.1. *Continuum powder models*

When modelled as a continuum, the effective thermal conductivity k_{eff} of a loose powder can be considered as the sum of three contributions [75]:

$$k_{\text{eff}} = k_r + k_s + k_g \tag{24}$$

Here k_r represents inter-particle radiation, k_s is the effective conductivity of the solid particle network and k_g represents conduction through the intervening gas (Fig. 4).

The radiative term can be approximated [84] as:

$$k_r = \frac{16}{3} L \sigma_B T^3 \tag{25}$$

Here T is temperature, σ_B is the Stefan-Boltzmann constant and L is the photon mean free path, which to a first approximation is equal to the particle diameter. The radiative term is very small, for example only 0.07 W m^{-1} K^{-1} for L = 40 µm even at a temperature of 1500 °C. Nevertheless, it is an important mode of heat transfer for electron beam melting since in a vacuum the k_g term becomes negligible.

In a loose powder, the rate of conduction through the solid material is known to be primarily limited by the size of the contacts between particles. Tolochko et al. [84] performed numerical calculations supported by experiment which gave the following estimate:

$$k_s = \frac{\Lambda a}{r} k_m \tag{26}$$

Here k_m is the metal bulk conductivity, r is the particle radius and a is the radius of connecting necks. The quantity Λ is a geometric factor to account for the degree of coordination of particles, increasing from a minimum of 0.4 for a diamond-like arrangement to a maximum of 1.7 for a BCC arrangement. The problem in practical application of eq. (26)

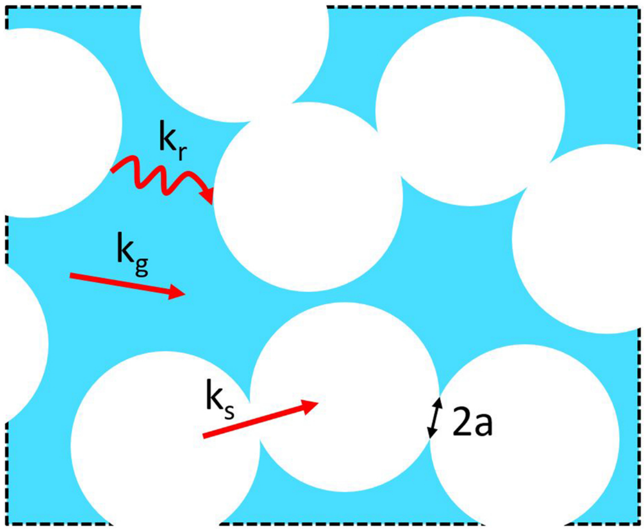

Fig. 4. Portion of a powder bed illustrating three contributions to the effective thermal conductivity.

is the uncertainty in the radius a of the inter-particle contacts. The contact radii are created by gravity-induced deformations [85] and will depend on particle shapes and orientation. For SLM they effectively start near zero but may be more significant for electron beam melting due to sintering produced by pre-heating.

The contribution k_g from gas conduction is also difficult to estimate, and the subject of many theoretical studies, as others have summarised [86–88]. A number of simplifying assumptions are required to even obtain a tractable expression. First it is assumed that the bed consists of spheres, and in a narrow size range. It is also assumed that direct particle–particle conduction is negligible and that the ratio of metal to gas conductivity, k_m/k_{gas}, is larger than about 1000. Note that k_{gas} refers to the gas bulk conductivity, not the *effective* conductivity k_g of gas in a pore network that we are seeking to estimate. The simplest expression for k_g that is applicable to solid volume fractions F_m in the range seen in additive manufacturing is due to Meredith and Tobias [89]:

$$k_g = \frac{(1 + F_m)(2 + F_m)}{(1 - F_m)(2 - F_m)} k_{\mathrm{gas}} \tag{27}$$

Note that in eq. (27) k_g includes the effect of conduction through the solid spheres, with a high diffusivity due to the large k_m/k_{gas} limiting value, but omits any sphere–sphere contact. The omission of particle–particle contact conduction justifies applying this expression for k_g in eq. (24).

To give a sense of the magnitudes of the various terms in eq. (27), consider a Ti-6Al-4 V bed at 1500 °C with a volume fraction $F_m = 0.6$. At this temperature, k_m for Ti-6Al-4 V is about 25 W m^{-1} K^{-1} and k_{gas} for argon gas is 0.063 W m^{-1} K^{-1}. Then eq. (27) gives an effective gas conductivity of $k_g = 0.47$ W m^{-1} K^{-1}. (Although the k_m/k_{gas} ratio of about 400 does not strictly meet the $k_m/k_{gas} > 1000$ requirement, it suffices for this rough estimate.)

The gas effective conductivity estimated by eq. (27) is independent of particle size due to the underlying assumptions. More refined treatments consider an additional gas heat transfer mechanism due to free molecular transport in regions where surfaces are closer than the gas mean free path. The mean free path can be calculated theoretically from thermodynamic quantities. Extremely complicated expressions for k_g ensue, such as the Zehner–Schlunder-Bauer (ZSB) model [86,88], which do have a particle-size dependence that may be correlated with observations. A few groups have attempted to incorporate these expressions in SLM simulations, albeit in simplified forms:

- Gu and co-workers at the Nanjing University of Aeronautics and Astronautics [53,59,90–92] simulated laser melting of an AlSi10 Mg powder. Their powder conductivity expression, as explained in earlier work [93], is a variant of the ZSB model for solid spheres surrounded by gas and was developed by Sih and Barlow [94]. In the Nanjing group's implementation, particle contacts are ignored because terms involving the flattening of contacts between spheres are omitted. Some other groups [54,95,96] have adopted the same method.
- Mukherjee et al. [97] apply an expression developed by Rombouts et al. (eq. 23 in [86]) for the limiting case of zero contact areas, but with an allowance for rapid gas conduction at the contact points. Unlike the Sih and Barlow expression, there is no radiative term.

However, the value of advanced theoretical treatments of powder conductivity is dubious for AM simulations:

- The real powder beds contain particles that are not exactly spherical and in a range of sizes. The inter-particle contact areas are unknown, highly variable but *finite*, especially since there will be some sintering of powder adjacent to a previous beam track, and due to pre-heating in the case of EBM.
- The earlier discussion shows that unsintered powder has very low effective conductivity, at most only a few percent of the bulk value. It is possible that the exact value used in the simulation is unimportant; this will depend on the conditions being modelled and can be investigated by sensitivity studies.
- The effective conductivity increases rapidly where the beam starts to heat and melt the powder. Some type of transition of the effective conductivity to the bulk value needs to be assumed, such as a step change or a ramp over a temperature interval. The form of transition could influence the overall results significantly. However, it is possible that the formation of a melt connection between particles happens so rapidly that through-melt conduction quickly dominates. Then the powder conductivity model can be decided mainly on the basis of numerical robustness.

In contrast to the above theoretical methods, a few authors have attempted to actually measure the effective conductivity of the powder bed to apply in their AM simulations:

- Neira Arce [98] presented data for two types of Ti-6Al-4 V powder at temperatures up to 1600 °C, finding that the values ramped up more steeply above about 700 °C.
- Cheng et al. [63] measured the thermal conductivity of Ti-6Al-4 V powder preserved after EBM preheating and found it increased from 0.63 W m^{-1} K^{-1} at room temperature to 2.44 at 750 °C.
- Andreotta et al. [99] measured the thermal conductivities of Inconel 718 powder beds at three packing densities from 300 K to 1000 K.
- Zhang et al. [100] found that the thermal conductivities of Ti-6Al-4 V and Inconel 625 powders increased from room temperature to 500 °C. However, both powders maintained approximately constant percentages of their bulk values over this temperature range, 4–5 % and 4–7 % respectively.
- Most recently, Wei et al. [101] measured the effective conductivities of five different powders over a range of gas pressures (He, Ar, N_2) and found that the gas contribution k_g dominates over the particle contact term k_s in all cases.
- Moser et al. [85] directly calculated the effective conductivity of a powder using a DEM heat transfer model. Spherical particles were loaded into a virtual box, a heat flux applied and the effective conductivity obtained from the resultant steady-state temperatures. Although subject to several approximations, their results are well-validated and show informative sensitivities to variables such as temperature, depth, gas conductivity, emissivity and packing density. The shortcoming is a lack of predictive ability as the model

would need to be run for each particular bed material.

Given the uncertainties and complications discussed earlier, it is perhaps not surprising that the authors of most melt pool simulations do not even describe their treatment of the powder conductivity. Possibly they simply apply the bulk metal value, raising doubts about the accuracy of their results. A quick summary of those publications which *do* mention their method is given below.

- For laser melting of Ti-6Al-4 V, Roberts et al. [62] quoted a text [102] suggesting that $k_{\mathrm{eff}} = F_m k_m$. They also reference the work of Rombouts et al. [86] and indicate that k_{eff} should rise from ~ 0.1 W m^{-1} K^{-1} at room temperature to ~ 0.3 W m^{-1} K^{-1} around the melting point. They did not state what values they actually implemented, but in another publication [103] mentioned they use a value of $k_{\mathrm{eff}} = 0.25$ W m^{-1} K^{-1}, presumably a constant. Verhaeghe et al. [66] used a fixed value of $k_{\mathrm{eff}} = 0.20$ W m^{-1} K^{-1} for the same process.
- For electron beam melting of Ti-6Al-4 V, Jamshidinia et al. [104] provided a graph and table of their k_{eff} values. They use a fixed value of 0.2 W m^{-1} K^{-1} up to 1000 °C and then a ramp up to the melt value of 34.6 W m^{-1} K^{-1}. The ramp was applied for numerical convergence reasons, not physical reasons. In fact the full melt value was not applied until several hundred degrees above the liquidus.
- For laser melting of a stainless steel, Gusarov et al. [77] used a fixed small conductivity below the melt temperature, and a step change to the melt value. They cited Rombouts et al. [86] regarding the gas contribution k_g and employed a value of 0.3 W m^{-1} K^{-1} calculated from gas conductivity just below the melt temperature and (presumably) eq. (27) or similar. They stated that the radiative contribution is negligible on the basis of eq. (25).
- Zah and Lutzmann [72] also modelled laser melting of stainless steel. They stated that k_{eff} values calculated from the ZSB model range from 0.03 to 0.1 W m^{-1} K^{-1} below the solidus temperature. However, they did not explain what conductivities were actually applied in their simulation. Both Zhang et al. [47] and Foroozmehr et al. [105] set the powder conductivity to 1% of the bulk value in their simulations of the same process.
- For laser melting of Ti-6Al-4 V, Zeng et al. [58] scaled their powder conductivity as 0.602 times the bulk value. However, this scaling factor is dubious since it was obtained from measurements of already-sintered powder [106].
- For a pure iron powder, Yin et al. [48] set $k_{\mathrm{eff}} = (F_m)^4 k_m$, although in the cited reference [107] the exponent is said to be an empirical parameter of 2–3, not 4. They assumed porosity was constant up to the solidus. Karayagiz et al. [11] applied the same method to a Ti-6Al-4 V powder.
- For electron beam melting of Ti-6Al-4 V, Shen and Chou [67] modelled the powder conductivity as the sum of the k_r and k_s terms in eqs (25) and (26). However, they did not say what neck radius or geometric factor Λ were assumed, or supply any actual powder conductivity values.
- Arisoy et al. [65,108] used $F_m k_m$ when simulating SLM of Inconel 625.
- In their simulation of AM by laser melting, Romano et al. [75] described the powder effective properties in general, including the Eq.s (24)–(27) provided earlier. They presented a graph of the powder conductivities they employed for Ti-6Al-4 V, SS316 and Al7075. The equation parameters used to calculate these curves were not provided. Their graph for Ti-6Al-4 V is very coarse but shows a conductivity < 1 W m^{-1} K^{-1} up to ~1000 °C and then ramping linearly to ~ 5 W m^{-1} K^{-1} by about 1600 °C.

Finally, it is noted that the powder density in the simulation must be scaled by the metal volume fraction F_m for correct mass and so that the enthalpy increase of the powder material matches the beam power absorbed. This is implemented in virtually all published continuum simulations. The difficulty arises when the powder subsequently consolidates to a melt. Ensuring mass, volume and enthalpy conservation across this transition can be problematic, depending on the numerical method employed. In some cases it is achieved by modification of cell fill fractions, or by element deactivations. However, in many publications the issue appears to be left unaddressed, which implies the likelihood of large errors.

2.5.2. Discrete powder models

On the face of it, treating the powder as discrete particles would seem to obviate all the conductivity problems. However, that is not the case. Most authors do not explicitly model the gas as a second phase, and so are relying on point–point particle contacts for heat transfer in the bed. This means the rate of heat transfer may be significantly underestimated. Furthermore, any effects of sintering are missed, and it is more difficult than with a continuum model to apply a workaround such as a spatially varying effective conductivity. Of the groups that have modelled discrete particles, only the following information about powder conductivity considerations has been published:

- Gurtler et al. [69] constructed a powder bed as a regular arrangement of spheres with bimodal sizes. They adjusted the separation of the spheres to control the density, which also affects overlaps and therefore thermal conductivity via contact areas. They stated that 'Despite the thermal conductivities of our powders at room temperature are overestimated due to geometrical constraints, we point out that their thermal conductivity at much higher process temperatures can safely be assumed to match the experimental values more closely'.
- The LLNL group simulated discrete particles but did not explain exactly how they handle powder conductivity, although some information is provided in a review by King et al. [3]. This mentions a code validation test in which they computed the thermal conductivity of stainless steel powder as spheres in air, and compared the ratio of effective conductivity to air for three packing densities and obtained reasonable agreement with data by Rombouts et al. [86]. It also appears that the LLNL code incorporates the thermal diffusive effect of gas between particles but not gas flow in general: 'The current powder model ignores the gas dynamics of any trapped background gas, instead modelling only the metal and a "void material"'.
- Description of the powder assumptions employed by researchers associated with ESI is also scarce. From the description in Megahed et al. [4], it seems that a gas phase is included in their simulation. However, it also seems there is no particle–particle contact, so heat transfer must be purely through the intervening gas until the particles start to melt.

3. Summary of published research

This section describes the published research in the field of melt pool simulation for additive manufacturing. In sections 3.1 to 3.4 respectively, research by the four most prominent groups is discussed. These groups have been selected, somewhat subjectively, on the basis of their simulation capabilities, number of authors and publication output. The work of the remaining groups is outlined in section 3.5, with fuller descriptions of some given in Appendix B.

A table of key information is given for each research group. For clarity and brevity, the simulation capability of each group is presented as *gaps* and *enhancements* relative to a fully-capable benchmark. This hypothetical benchmark combines all the features of the current leading groups:

- 3D geometry with powder bed as discrete particles
- Heat transfer including evaporative and radiative losses